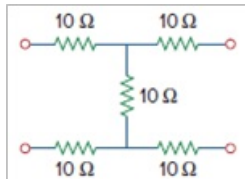


Problem

Obtain the z parameters for the network in Fig. 19.65.

Figure 19.65

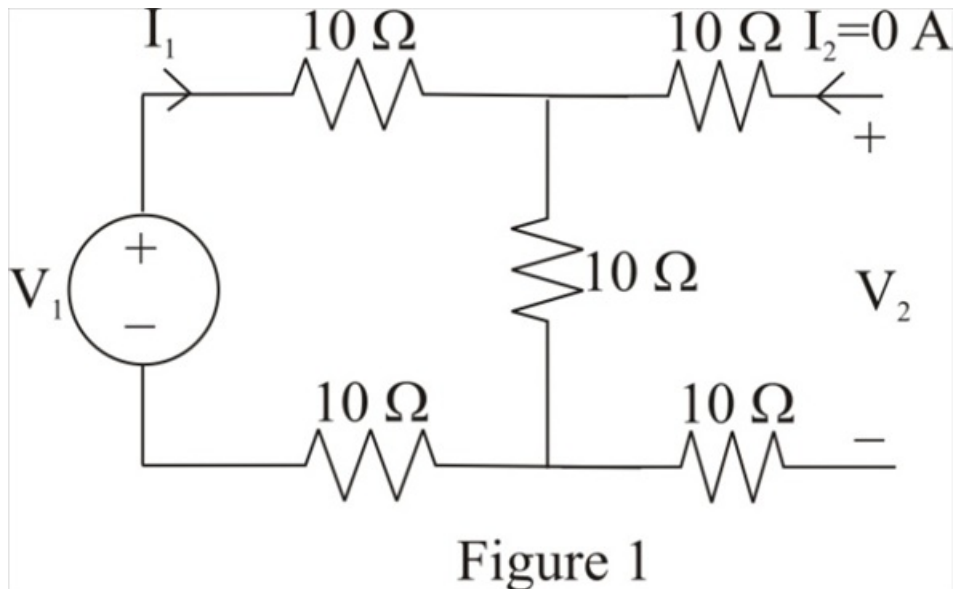


Step-by-step solution

Step 1 of 8

Refer to Figure 19.65 in the text book.

Connect the voltage source V_1 as shown in Figure 1. Identify the currents and voltages in the circuit.



Step 2 of 8

Consider the expression for the voltage V_1 .

$$\begin{aligned} V_1 &= I_1(10 + 10 + 10) \\ &= I_1(30) \end{aligned}$$

Therefore, the expression for the voltage V_1 is $I_1(30)$.

Consider the expression for the voltage V_2 .

$$\begin{aligned} V_2 &= I_1(10) \\ &= I_1(10) \end{aligned}$$

Therefore, the expression for the voltage V_2 is $I_1(10)$.

Step 3 of 8

Determine the expression for the open circuit input impedance z_{11} .

$$z_{11} = \frac{V_1}{I_1}$$

Substitute $I_1(30)$ for V_1 .

$$\begin{aligned} z_{11} &= \frac{I_1(30)}{I_1} \\ &= 30 \, \Omega \end{aligned}$$

Therefore, the expression for the open circuit input impedance z_{11} is $30 \, \Omega$.

Step 4 of 8

Determine the value of the open circuit transfer impedance z_{21} from port-2 to port-1.

$$z_{21} = \frac{V_2}{I_1}$$

Substitute $I_1(10)$ for V_2 .

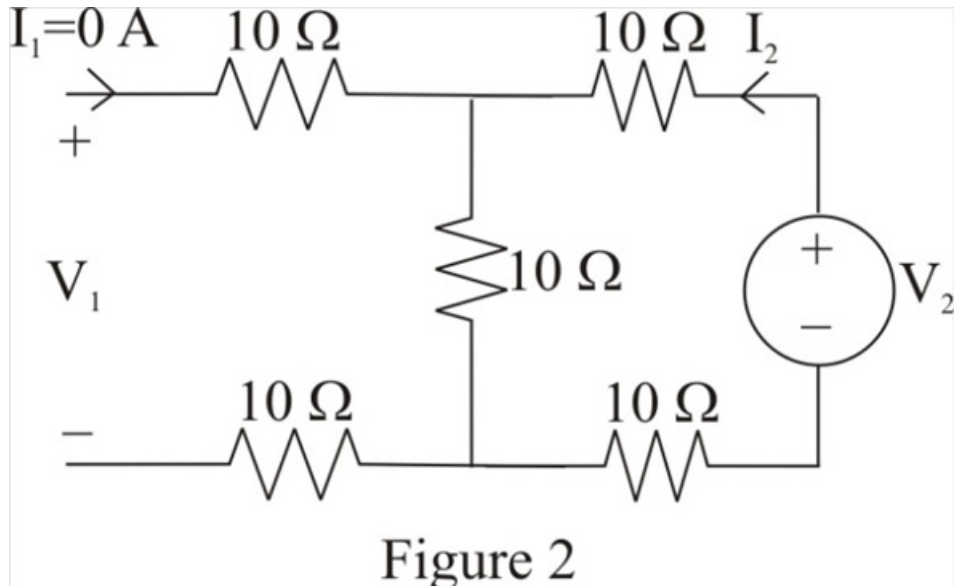
$$\begin{aligned} z_{21} &= \frac{I_1(10)}{I_1} \\ &= 10 \, \Omega \end{aligned}$$

Therefore, the value of the open circuit transfer impedance z_{21} from port-2 to port-1 is

$$10 \, \Omega.$$

Step 5 of 8

Connect the voltage source V_2 as shown in Figure 2. Identify the currents and voltages in the circuit.



Step 6 of 8

Consider the expression for the voltage V_2 .

$$\begin{aligned} V_2 &= I_2(10+10+10) \\ &= I_2(30) \end{aligned}$$

Therefore, the expression for the voltage V_2 is $I_2(30)$.

Consider the expression for the voltage V_1 .

$$V_1 = I_2(10)$$

Therefore, the expression for the voltage V_1 is $I_2(10)$.

Step 7 of 8

Determine the value of the open circuit transfer impedance z_{12} from port-1 to port-2.

$$z_{12} = \frac{V_1}{I_2}$$

Substitute $I_2(10)$ for V_1 .

$$\begin{aligned} z_{12} &= \frac{I_2(10)}{I_2} \\ &= 10 \Omega \end{aligned}$$

Therefore, the value of the open circuit transfer impedance z_{12} from port-1 to port-2 is

$$\boxed{10 \Omega}.$$

Step 8 of 8

Determine the expression for the open circuit output impedance z_{22} .

$$z_{22} = \frac{V_2}{I_2}$$

Substitute $I_2(30)$ for V_2 .

$$\begin{aligned} z_{22} &= \frac{I_2(30)}{I_2} \\ &= 30 \Omega \end{aligned}$$

Therefore, the expression for the open circuit output impedance z_{22} is $\boxed{30 \Omega}$.